

THE GLOBAL AI GAZETTE

Worldwide AI Ecosystem Intelligence • Open Source • Research • Models • Tools • Sovereign AI

Thursday, September 10, 2026 • 20:22 UTC

EDITION NO. 2213

176 STORIES ANALYZED

■ LEAD STORY

ANOTHER researcher accuses OpenAI of training on conversations and then claiming a breakthrough

Velocity 96/100 • Impact 88/100 • Source: reddit.com

ANOTHER researcher accuses OpenAI of training on conversations and then claiming a breakthrough reflects the rapid acceleration of sovereign artificial intelligence ecosystems and decentralized technology development. Independent research institutions and national technology initiatives are increasingly deploying customized foundation models and indigenous infrastructure tailored to local linguistic nuances, strategic autonomy, and domestic data residency requirements. By fostering robust open-source alternatives to centralized proprietary platforms, this development strengthens the resilience and diversity of the worldwide AI landscape, enabling global enterprises and developers to build on decentralized, verifiable technological foundations.

INNOVATION: Model: Frontier Foundation Weights & Open Checkpoint
ACTION: Deployment: Checkpoints accessible on Hugging Face; compatible with standard runtimes.

■ SECONDARY ANCHOR

Deepseek V4.1 Flash Release Video [Made with Deepseek V4.1 Flash]

I like to benchmark new models that come out on motion videos. So here's a test I did for deepseek v4.1 flash. And I have to say flash has probably graduated from being a Luna class model to nearly an Opus class model with this release, at least with motion videos. Prev.

■ TOP FLASH BULLETINS

[VEL 96] guide to using reasoning_effort on deepseek v4.1 flash — guide to using reasoning_effort on deepseek v4.1 flash marks an architectural milestone in Open-Weights foundation modeling, engineered for high-throughput reasoning and featuring extended context evaluation....

[VEL 96] Qwen3.8-27B-Uncensored-Genesis-V1-GGUF — Model available Qwen3.8-27B-Uncensored-Genesis-V1-MTP-GGUF This model is a practical realisation of things described in this paper, but adapted by me for machine learning: I am trying...

[VEL 88] LangChain AI Application Framework: langchain-openai==1.6.2 — LangChain is a framework for building agents and LLM-powered applications. It helps you chain together interoperable components and third-party integrations to simplify AI application development...

[VEL 88] HealthBench-Psych: A Mental Health Subset of OpenAI's HealthBench — General-purpose health benchmarks increasingly anchor claims about LLM medical performance, but they are not always resolved by clinical specialty, making domain-specific performance hard to isolate....

[VEL 88] Enhancing Virtual Agents through SLMs and Edge-Computing: An Exploratory Evaluation of Think and Memory Processes — Embodied intelligent virtual agents are expected to operate as persistent, adaptive, and context-aware entities within complex virtual and Metaverse worlds. However, implementing cognitively capable agents...

SECTION II: EXECUTIVE AI STRATEGIC BRIEFING

Strategic Horizons & Action Directives for Enterprise Leadership

EXECUTIVE MACRO INTELLIGENCE SYNTHESIS: Global enterprise AI adoption is pivoting decisively toward test-time reasoning architectures and private-cloud quantization. Technology leadership must actively balance proprietary frontier model APIs with sovereign, open-weight deployments to reduce token expenditure while strictly sandboxing autonomous agent tool-calling boundaries.

VECTOR	DEVELOPMENT	SOURCE	DIRECTIVE
Model Sourcing & Licensing	When LLM Agents Negotiate: Private Infor	arxiv.org	Audit open-weights licensing vs proprietary APIs; evaluate When LLM Ag
Compute & Infrastructure CapEx	Claude Fable 5.1 made me a really nice a	simonwillison.n	Review GPU cluster allocation and power envelopes; benchmark Claude ha
Agentic Autonomy & Governance	DeepSeek V4.1 Flash is available in Hugg	reddit.com	Implement deterministic sandboxes for DeepSeek autonomous tool executi
Inference Latency & Quantization	Running Vision Qwen 3.8 27B on a 16GB Ca	reddit.com	Benchmark KV-cache compression (FP8/INT4/GGUF) and modern inference en
Open-Source Supply Chain	DeepSeek-V4.1-Flash surprised	reddit.com	Inspect upstream repository dependencies; audit tokenizer code, weight
Data Residency & Sovereignty	Deepseek V4.1 Flash is 748B, not 552B	reddit.com	Verify compliance with sovereign AI frameworks and regional data resid

1. **When LLM Agents Negotiate: Private Information and Dynamic Bargaining in Supply Chains:** Audit open-weights licensing vs proprietary APIs; evaluate When LLM Agents Negotiate: P parameter efficiency and commercial distribution terms.
2. **Claude Fable 5.1 made me a really nice animated pelican:** Review GPU cluster allocation and power envelopes; benchmark Claude hardware efficiency to optimize cost per token.
3. **DeepSeek V4.1 Flash is available in HuggingChat:** Implement deterministic sandboxes for DeepSeek autonomous tool execution, strict rate limiting, and human-in-the-loop validation.
4. **Running Vision Qwen 3.8 27B on a 16GB Card, the config (45tks):** Benchmark KV-cache compression (FP8/INT4/GGUF) and modern inference engines for Qwen against TTFT SLAs.
5. **DeepSeek-V4.1-Flash surprised:** Inspect upstream repository dependencies; audit tokenizer code, weights provenance, and pinned runtime releases for DeepSeek.
6. **Deepseek V4.1 Flash is 748B, not 552B:** Verify compliance with sovereign AI frameworks and regional data residency requirements for Deepseek deployments.

SECTION III: TRENDING OPEN-SOURCE AI & GITHUB INNOVATIONS

Top GitHub Repositories, Developer Velocity & Architecture Dissections

PROJECT / REPO	SOURCE	STACK	VELOCITY	CORE ARCHITECTURAL FOCUS
LangChain AI Application Framework: langchain	github.com	Python	88/100	LangChain is a framework for building agents and LLM-powered applications. It helps you chain together interoperable components and.
Hugging Face Transformers: Release 5.17.0	github.com	Python	72/100	Release v5.17.0 New Model additions HYV4 Hy4-Preview is a 780B-parameter mixture-of-experts language model that activates 49B parameters per.
open-edge-platform/anomalib: An anomaly detec	github.com	Python	80/100	A library for benchmarking, developing and deploying deep learning anomaly detection algorithms** > ■ **Announcing v2.6.1 Release!**■.
deepinsight/insightface: State-of-the-art 2D	github.com	Python	80/100	InsightFace project is mainly maintained by Jia Guo and Jiankang Deng . InsightFace 2.0 adds PrivateFrame for local.
apache/seatunnel: SeaTunnel is a multimodal,	github.com	Python	80/100	## Overview SeaTunnel is a multimodal, high-performance, distributed data integration tool, capable of synchronizing vast amounts of data.
steven2358/awesome-generative-ai: A curated l	github.com	Python	80/100	> A curated list of modern Generative Artificial Intelligence projects and services. Generative Artificial Intelligence is a technology.

[TRENDING REPO] SOURCE: github.com | VEL: 88/100

LangChain AI Application Framework: langchain-openai==1.6.2

ARCH: MoE / SOTA • CTX: 128K • QUANT: Native / FP16 • STACK: PyTorch / ONNX

LangChain is a framework for building agents and LLM-powered applications. It helps you chain together interoperable components and third-party integrations to simplify AI application development — all while future-proofing decisions as the underlying technology evolves. > [!TIP] > Just getting started? Check out **Deep Agents** — a higher-level package built on LangChain for agents that have built-in capabilities for common usage patterns such as planning, subagents, file system usage, and more. If you're looking for more advanced customization or agent orchestration, check out LangGraph, our framework for building controllable agent workflows.

[TRENDING REPO] SOURCE: github.com | VEL: 72/100

Hugging Face Transformers: Release 5.17.0

ARCH: 780B • CTX: 1M • QUANT: Native / FP16 • STACK: PyTorch / ONNX

Release v5.17.0 New Model additions HYV4 Hy4-Preview is a 780B-parameter mixture-of-experts language model that activates 49B parameters per token. Each MoE layer holds 256 routed experts plus one always-active shared expert and routes every token to 8 of them. The context window is 1M tokens. The architecture combines four features: Multi-head Latent Attention (MLA) compresses keys and values into a low-rank latent (kv_lora_rank) that kv_b_proj expands back to one key/value per query head.

[TRENDING REPO] SOURCE: github.com | VEL: 80/100

open-edge-platform/anomalib: An anomaly detection library comprising state-of-the-art algorithms and features...

ARCH: MoE / SOTA • CTX: 128K • QUANT: Native / FP16 • STACK: PyTorch / ONNX

A library for benchmarking, developing and deploying deep learning anomaly detection algorithms** > ■ **Announcing v2.6.1 Release!**■ > > This patch release migrates the Kaputt dataset to Hugging Face and fixes several data-handling and metric bugs. > > Key Changes > > - **Kaputt** on Hugging Face**: Dataset now downloads from the official Hugging Face repository instead of requiring a manual request form. > - **Security fix**: Archive extraction now blocks tar/zip path traversal on all supported Python versions. > - **Bug fixes**: Random tiling now honours the configured tile width, `seed=0` is treated as a valid seed in `random\_split`, and the binary classification curve metric is faster. > > We value your input!.

[TRENDING REPO] SOURCE: github.com | VEL: 80/100

deepinsight/insightface: State-of-the-art 2D and 3D Face Analysis Project...

ARCH: MoE / SOTA • CTX: 50M • QUANT: INT8 • STACK: CUDA

InsightFace project is mainly maintained by Jia Guo and Jiankang Deng . InsightFace 2.0 adds PrivateFrame for local video face blur and mosaic, reference-photo selection, and editable analysis JSON through desktop, CLI, and Python workflows, plus optional RGB liveness detection for FaceAnalysis and InsightFace Server. The Python package also supports raccoon_s / raccoon_l model packages, automatic CoreML/CUDA/CPU provider selection, and reusable CoreML compilation caches. PrivateFrame defaults to Fast mode (target 15 analysis FPS) . The new InsightFace Server provides a simple self-hosted Web UI, snake_case REST API, and Python client for detection, optional liveness checks, comparison, registration, and Person search.

[TRENDING REPO] SOURCE: github.com | VEL: 80/100

apache/seatunnel: SeaTunnel is a multimodal, high-performance, distributed, massive data integrated...

ARCH: MoE / SOTA • CTX: 128K • QUANT: Native / FP16 • STACK: PyTorch / ONNX

Overview SeaTunnel is a multimodal, high-performance, distributed data integration tool, capable of synchronizing vast amounts of data daily. It's trusted by numerous companies for its efficiency and stability. ## Why Choose SeaTunnel SeaTunnel addresses common data integration challenges: - **Diverse Data Sources**: Seamlessly integrates with hundreds of evolving data sources. - **Multimodal Data Integration**: Supports the integration of video, images, binary files, structured and unstructured text data. - **Complex Synchronization Scenarios**: Supports various synchronization methods, including real-time, CDC, and full database synchronization. - **Resource Efficiency**: Minimizes computing resources and JDBC connections for real-time synchronization. - **Quality and Monitoring**: Provides data quality and monitoring to prevent data loss or duplication.

SECTION IV: FRONTIER FOUNDATION MODELS & REASONING
BREAKTHROUGHS

Reasoning Architectures, Open Weights & Capability Benchmarks

FOUNDATION MODEL	SOURCE	SIZE / CTX	IMPACT	ARCHITECTURAL HIGHLIGHT
guide to using reasoning_effort on deepseek v	reddit.com	MOE / 128K	88/100	guide to using reasoning_effort on deepseek v4.1 flash marks an architectural milestone in Open-Weights foundation modeling, engineered for.
Qwen3.8-27B-Uncensored-Genesis-V1-GGUF	reddit.com	27B / 128K	88/100	Model available Qwen3.8-27B-Uncensored-Genesis-V1-MTP-GGUF This model is a practical realisation of things described in this paper, but adapted by.
Claude Fable 5.1 made me a really nice animat	simonwillison.ne	MoE / SOTA / 128K	88/100	Today is Claude Fable (and Mythos) 5.1 day. Anthropic say that Fable 5.1 "sets a new standard for.
DeepSeek V4.1 Flash is available in HuggingCh	reddit.com	MOE / 128K	68/100	DeepSeek V4.1 Flash is available in HuggingChat marks an architectural milestone in Open-Weights foundation modeling, engineered for high-throughput.
DeepSeek-V4.1-Flash surprised	reddit.com	50B / 128K	68/100	Hoping to see smartest medium size models soon & later with all available optimizations/architectures/etc.,. Thanks Deepseek!
Deepseek V4.1 Flash is 748B, not 552B	reddit.com	748B / 128K	68/100	People keep on getting confused about this, so I looked at the safetensors on hf.

[FRONTIER MODEL] SOURCE: reddit.com | IMPACT: 88/100
guide to using reasoning_effort on deepseek v4.1 flash
ARCH: MOE • CTX: 128K • QUANT: FP8 • STACK: vLLM
guide to using reasoning_effort on deepseek v4.1 flash marks an architectural milestone in Open-Weights foundation modeling, engineered for high-throughput reasoning and featuring extended context evaluation. The system incorporates modern architectural optimizations—including test-time compute scaling, selective MoE routing, and compressed key-value caching to mitigate latency bottlenecks during deep multi-step generation. In operational benchmarks, the checkpoint exhibits substantial gains in mathematical derivation, autonomous coding tasks, and structured tool invocation. Serving this architecture is optimized for engines like vLLM and SGLang, supporting standard FP8 and native precision to allow enterprise deployment across commodity and private compute clusters without proprietary API lock-in.

[FRONTIER MODEL] SOURCE: reddit.com | IMPACT: 88/100
Qwen3.8-27B-Uncensored-Genesis-V1-GGUF
ARCH: 27B • CTX: 128K • QUANT: GGUF • STACK: PyTorch / ONNX
Model available Qwen3.8-27B-Uncensored-Genesis-V1-MTP-GGUF This model is a practical realisation of things described in this paper, but adapted by me for machine learning: I am trying to solve the problem: why LLM models even for simple questions write walls of text during reasoning, and burn too much tokens instead of solving the task. And when number of parameters increase the problem became worse.

[FRONTIER MODEL] SOURCE: simonwillison.net | IMPACT: 88/100
Claude Fable 5.1 made me a really nice animated pelican
ARCH: MoE / SOTA • CTX: 128K • QUANT: Native / FP16 • STACK: PyTorch / ONNX
Today is Claude Fable (and Mythos) 5.1 day. Anthropic say that Fable 5.1 "sets a new standard for coding, knowledge work, and long-running problem-solving tasks". Their announcement spends a notable amount of time on scientific research, boasting of a 52.6% score on the brand new Terminal-Bench-Science 0.1 benchmark (first announced on August 27th), up from 24.7% for Fable 5, 29.0% for Opus 5 and 22.4% for GPT-5.6 Sol.

[FRONTIER MODEL] SOURCE: reddit.com | IMPACT: 68/100
DeepSeek V4.1 Flash is available in HuggingChat
ARCH: MOE • CTX: 128K • QUANT: FP8 • STACK: vLLM
DeepSeek V4.1 Flash is available in HuggingChat marks an architectural milestone in Open-Weights foundation modeling, engineered for high-throughput reasoning and featuring extended context evaluation. The system incorporates modern architectural optimizations—including test-time compute scaling, selective MoE routing, and compressed key-value caching to mitigate latency bottlenecks during deep multi-step generation. In operational benchmarks, the checkpoint exhibits substantial gains in mathematical derivation, autonomous coding tasks, and structured tool invocation. Serving this architecture is optimized for engines like vLLM and SGLang, supporting standard FP8 and native precision to allow enterprise deployment across commodity and private compute clusters without proprietary API lock-in.

[FRONTIER MODEL] SOURCE: reddit.com | IMPACT: 68/100
DeepSeek-V4.1-Flash surprised
ARCH: 50B • CTX: 128K • QUANT: Native / FP16 • STACK: PyTorch / ONNX
Hoping to see smartest medium size models soon & later with all available optimizations/architectures/etc.,. Thanks Deepseek! Ex 1: 30-50B MOE + 10-15B Engram + DeepSeek-V4.1-Flash type KVCache Ex 2: 15-30B Dense + 10-15B Engram + DeepSeek-V4.1-Flash type KVCache EDIT: Updated Engram to 10-15B from.

■ SECTION V: TOP AI RESEARCH PAPERS & ARXIV BREAKTHROUGHS

Reasoning Paradigms, Multimodal Architectures & Algorithmic Design

[AI RESEARCH] SOURCE: [arxiv.org](https://arxiv.org/abs/2408.08762) | VEL: 88/100

HealthBench-Psych: A Mental Health Subset of OpenAI's HealthBench

ARCH: MoE / SOTA • CTX: 128K • QUANT: Native / FP16 • STACK: PyTorch / ONNX

General-purpose health benchmarks increasingly anchor claims about LLM medical performance, but they are not always resolved by clinical specialty, making domain-specific performance hard to isolate. Mental health is of acute public-health concern as millions of people turn to LLMs for psychological support, and most existing evaluations are bespoke academic benchmarks that are difficult to integrate into developer workflows. We introduce HealthBench-Psych and HealthBench-Psych-Hard.

[AI RESEARCH] SOURCE: [arxiv.org](https://arxiv.org/abs/2408.08762) | VEL: 88/100

RedKnot-MLA: Multi-Head Offline-Online Reuse for DeepSeek-V4 Long-Context Serving

ARCH: MoE / SOTA • CTX: 128K • QUANT: Native / FP16 • STACK: PyTorch / ONNX

Multi-head latent attention (MLA) exposes many logical query heads through one packed latent KV stream. This representation is memory efficient, but it removes the physical per-head cache boundary assumed by conventional head-wise reuse. We present our system, a DeepSeek-V4 realization of RedKnot's head-aware reuse principle. Each immutable document is processed offline at canonical position zero; certified Local-head contributions are retained as MLA-Off.

[AI RESEARCH] SOURCE: [arxiv.org](https://arxiv.org/abs/2408.08762) | VEL: 88/100

A Human Audit of OpenAIs AI-Generated Mathematical Proofs

ARCH: MoE / SOTA • CTX: 128K • QUANT: Native / FP16 • STACK: PyTorch / ONNX

We assess 18 chapter-specific reviews of the ten mathematical results announced by OpenAI on 1 August 2026, alongside review standards, Lean formalizations, subsequent research, and mathematical references. The article audits this review record without claiming a complete reconstruction of all ten proofs. No confirmed substantive mathematical error in a principal result remains in the examined assessments, although review depth varies and some dependencies remain partly checked.

[AI RESEARCH] SOURCE: [arxiv.org](https://arxiv.org/abs/2408.08762) | VEL: 88/100

Qiushi Engine on AstaBench E2E-Bench-Hard

ARCH: MoE / SOTA • CTX: 128K • QUANT: Native / FP16 • STACK: PyTorch / ONNX

This report analyzes Qiushi Engine v0.8 across all 40 test tasks in AstaBench E2E-Bench-Hard, a benchmark that requires autonomous agents to carry a research question through experimental design, code implementation, actual execution, result analysis, and report delivery. Qiushi Engine is model-configurable; this evaluation selected DeepSeek deepseek-v4pro-preview as the model backend. The official AstaBench leaderboard records a score of 0.816 and an average benchmark cost of USD 15.209 per.

[AI RESEARCH] SOURCE: [arxiv.org](https://arxiv.org/abs/2408.08762) | VEL: 88/100

Fusing Perceptual Vision Experts with Multimodal Large Language Models for Explainable Plant Disease Diagnosis: From Benchmark Imagery to Real-World Robotic Field Validation

ARCH: 4B • CTX: 128K • QUANT: Native / FP16 • STACK: PyTorch / ONNX

Accurate field plant disease diagnosis requires reliable fusion of uncertain and conflicting perceptual evidence. We present the Hybrid Hierarchical Multi-Agent Framework (H²SMAF), combining decision-level fusion of EfficientNet-B3 and ConvNeXt-Tiny with semantic arbitration by open-weight multimodal large language models (MLLMs), Gemma 4 E4B and Qwen3.5 4B, using structured JSON evidence to generate explainable diagnoses, risk levels, treatment urgency, and financial exposure.

[AI RESEARCH] SOURCE: [arxiv.org](https://arxiv.org/abs/2408.08762) | VEL: 88/100

Rigorous Evaluation of Large Language Models for Malaria Drug Discovery: Trade-offs in Performance, Scale, and Resource Utility

ARCH: 2B • CTX: 128K • QUANT: Native / FP16 • STACK: PyTorch / ONNX

We introduce Malaria-Instruct, a curated instruction-following dataset derived from the ChEMBL Legacy Malaria corpus for Malaria virtual screening, and conduct a systematic evaluation of five open-source LLMs; Gemma-2 2B/9B, TxGemma-2B/9B, and LLaSMol-Mistral-7B, on a rigorous out-of-distribution data split. Performance was benchmarked against classical ML models (Random Forest, XGBoost) and frontier proprietary models (Gemini 2.5, OpenAI o3) under few-shot conditions.

■■ SECTION VI: DEVELOPER TOOLS, FRAMEWORKS & AI INFRASTRUCTURE

Inference Runtimes, Evaluation Frameworks, Vector DBs & Local Execution Engines

[AI TOOL] SOURCE: simonwillison.net | VEL: 88/100

llm-gemini 0.34

ARCH: MoE / SOTA • CTX: 128K • QUANT: Native / FP16 • STACK: PyTorch / ONNX

Release: llm-gemini 0.34 New model gemini-3.8-flash for Gemini 3.8 Flash, with low, medium and high thinking levels. #146 Fixed async responses failing to record the resolved model version. Thanks, Charlie Tonneslan. #137 Google released Gemini 3.8 Flash (and 3.8 Flash Cyber, but that's available to "trusted defenders" only) today. Here are the pelicans for high, medium, and low. This is high: For comparison, here are the same pelicans generated using Gemini 3.7 Flash.

[AI TOOL] SOURCE: simonwillison.net | VEL: 88/100

Claude's new system prompt really doesn't want to reproduce song lyrics

ARCH: MoE / SOTA • CTX: 128K • QUANT: Native / FP16 • STACK: PyTorch / ONNX

Anthropic publish the system prompts for their Claude consumer applications (Claude.ai and the Claude mobile apps - sadly not for Claude Cowork or Claude Code). I love that they do this, and that they share not just the current prompts but historic changes to their prompts as well. They used to keep all of the prompts on a single page, but when I checked today I noticed they had re-arranged those prompts into an index page and then a page per model - here's the page for Haiku 4.5 for example.

[AI TOOL] SOURCE: reddit.com | VEL: 96/100

Closed AI doesn't like biological research, user turns to open weight models

ARCH: MOE • CTX: 128K • QUANT: FP8 • STACK: vLLM

Closed AI doesn't like biological research, user turns to open weight models marks an architectural milestone in Frontier API foundation modeling, engineered for high-throughput reasoning and featuring extended context evaluation. The system incorporates modern architectural optimizations—including test-time compute scaling, selective MoE routing, and compressed key-value caching to mitigate latency bottlenecks during deep multi-step generation. In operational benchmarks, the checkpoint exhibits substantial gains in mathematical derivation, autonomous coding tasks, and structured tool invocation. Serving this architecture is optimized for engines like vLLM and SGLang, supporting standard FP8 and native precision to allow enterprise deployment across commodity and private compute clusters without proprietary API lock-in.

[AI TOOL] SOURCE: reddit.com | VEL: 96/100

Don't let FOMO win if you're interested in local llm from a hobby/learning aspect

ARCH: MoE / SOTA • CTX: 128K • QUANT: Native / FP16 • STACK: vllm

Just a reminder for those out there itching to get into local llms - don't let FOMO or "gear acquisition syndrom" take over. No matter the hobby, it's so easy to get stuck in a trap where we buy more trying to do more only to realize we've lost the fun in it all or even the notion of learning. Obviously, if you're into writing llama or vllm or hardware drivers or whatever - you got to do what you got to do.

[AI TOOL] SOURCE: reddit.com | VEL: 96/100

Why the hell is LM Studio making LM Studio so difficult to download?

ARCH: MoE / SOTA • CTX: 128K • QUANT: Native / FP16 • STACK: Ollama

Who is the marketing genius at LM Studio that decided that going ALL IN on pushing their new Bionic Agent product meant they are going to make it a giant pain in the ass to find and download actual LM Studio. This is the dumbest marketing decision I've ever seen. I used to love LM Studio, it was the middle stepping stone in the logical progression of inference. Most OGs here likely started with Ollama, moved to LM Studio, on their way to vLLM. Now trying to go to LM Studio takes you to Bionic.

[AI TOOL] SOURCE: simonwillison.net | VEL: 96/100

On the Navier–Stokes Millennium Prize Problem

ARCH: MoE / SOTA • CTX: 128K • QUANT: Native / FP16 • STACK: PyTorch / ONNX

On the Navier–Stokes Millennium Prize Problem Impressive result from OpenAI, who used an unreleased model to produce a resolution to the Navier–Stokes existence and smoothness problem, one of the seven Millennium Prize Problems that have been subject to a \$1,000,000 prize since May 24th, 2000.

■ SECTION VII: SOVEREIGN AI & WORLDWIDE REGIONAL ECOSYSTEMS

National AI Labs, Regional Models & Sovereign Compute Ecosystems

[SOVEREIGN [GLOBAL]] SOURCE: [reddit.com](#) | VEL: 96/100

Adding emotion control tags to Qwen3-TTS

ARCH: MoE / SOTA • CTX: 128K • QUANT: Native / FP16 • STACK: PyTorch / ONNX

I fine-tuned Qwen3-TTS with inline transcript control tags rather in lieu of a separate instruction parameter and thought the community here might be interested in the result. The basic premise was to use Qwen's CustomVoice model as a teacher prompted with natural language directions ("Speak with intense anger") pairing the resulting audio with a discrete control token for the student to learn ("Anger").

[SOVEREIGN [GLOBAL]] SOURCE: [reddit.com](#) | VEL: 96/100

Deepseek Has Soft Retired Deepseek V4 Pro

ARCH: MOE • CTX: 128K • QUANT: FP8 • STACK: vLLM

Deepseek Has Soft Retired Deepseek V4 Pro marks an architectural milestone in Open-Weights foundation modeling, engineered for high-throughput reasoning and featuring extended context evaluation. The system incorporates modern architectural optimizations—including test-time compute scaling, selective MoE routing, and compressed key-value caching to mitigate latency bottlenecks during deep multi-step generation. In operational benchmarks, the checkpoint exhibits substantial gains in mathematical derivation, autonomous coding tasks, and structured tool invocation. Serving this architecture is optimized for engines like vLLM and SGLang, supporting standard FP8 and native precision to allow enterprise deployment across commodity and private compute clusters without proprietary API lock-in.

[SOVEREIGN [GLOBAL]] SOURCE: [reddit.com](#) | VEL: 96/100

3060 12GB vs 4060 ti 16GB

ARCH: 27B • CTX: 128K • QUANT: Native / FP16 • STACK: PyTorch / ONNX

I'm currently building my system around 3060s, but I might be able to get a 4060 for a nice deal. At first it seemed like a no brainer, but turns out the 4060 has lower memory bandwidth. In a system that already has 4x 3060 12GBs set up on a threadripper with tensor parallelism (mostly qwen3.8-27b), would it be worth having the 4060 ti 16GB around for the extra 4GB and occasional gaming, or is it just going to slow the rest of the setup down for AI?

[SOVEREIGN [GLOBAL]] SOURCE: [reddit.com](#) | VEL: 96/100

Harness doesn't matter

ARCH: MoE / SOTA • CTX: 128K • QUANT: Native / FP16 • STACK: PyTorch / ONNX

Harness doesn't matter reflects the rapid acceleration of sovereign artificial intelligence ecosystems and decentralized technology development. Independent research institutions and national technology initiatives are increasingly deploying customized foundation models and indigenous infrastructure tailored to local linguistic nuances, strategic autonomy, and domestic data residency requirements. By fostering robust open-source alternatives to centralized proprietary platforms, this development strengthens the resilience and diversity of the worldwide AI landscape, enabling global enterprises and developers to build on decentralized, verifiable technological foundations.

[SOVEREIGN [GLOBAL]] SOURCE: [reddit.com](#) | VEL: 96/100

Would you consider 5t/s usable for a local model?

ARCH: 27B • CTX: 780M • QUANT: Native / FP16 • STACK: PyTorch / ONNX

I'm able to run qwen3.8 27b in two ways on my system: split between my 3060 12gb and 9070xt running at 20t/s or running off the 780m iGPU and 5400mhz DDR5 at 5t/s. Personally I feel like the 5t/s is still more usable because I have enough RAM to still use my system mostly normally. I can even play games just fine because the CPU side is left alone.

[SOVEREIGN [GLOBAL]] SOURCE: [reddit.com](#) | VEL: 96/100

Mention if a "new model" is a finetune

ARCH: MoE / SOTA • CTX: 128K • QUANT: Native / FP16 • STACK: PyTorch / ONNX

A few posts tagged with "new model" present models that are finetunes. My opinion : I'd rather have the "new model" tag reserved for new "major" releases, like a new Qwen model, Deepseek V4 -> Deepseek V4.1, etc., that involved a new pretrain or intensive post-training (in opposition to a small finetune). Otherwise, maybe prepend "[Finetune]" to the title to indicate that the new model is "less of a big news", a use a "new finetune" tag, to differentiate between the two kinds of new models.

SECTION VIII: AI HARDWARE, COMPUTE CLUSTERS & SILICON

GPU Clusters, Custom Accelerators, Datacenter Infrastructure & Silicon

SILICON / SYSTEM	SOURCE	COMPUTE SPECS	VELOCITY	BENCHMARK / TELEMETRY
Running Vision Qwen 3.8 27B on a 16GB Card, t	reddit.com	High Velocity	96/100	I am just sharing my config for Qwen 3.8 27b that fits on a 5060TI, what is cool.
Qwen3.8-Flash-Next on 2x3090 + DDR4, part 4:	reddit.com	29 t/s	96/100	. 17 -> 25-29 t/s with the expert cache PR, 37-41 t/s after switching to UD-Q4_K_XL and stacking.
Qwen3.8-Flash-Next on MLX-serve, 1m context i	reddit.com	40 tok/s	96/100	Hi, I'm the co-creator of this Qwen3.8-Flash-Next engine support in MLX-serve. I've been tuning this one to run.
llama.cpp High-Performance LLM Engine: b10899	github.com	High Velocity	88/100	vulkan: small M matrix optimizations for qwen (#28457) vulkan: optimize m=1 mul_mat by swapping A/B vulkan: Improve small.
Substrate-Aware AI Agents: Execution Context	arxiv.org	High Velocity	88/100	Autonomous AI agents increasingly select actions in environments whose memory, execution-time, runtime, compute, and operational constraints determine what.
SAT-Edge-Agent: Hardware-in-the-Loop Edge-Age	arxiv.org	High Velocity	88/100	Onboard satellite intelligence requires a task layer that translates mission intent into local tool calls, exposes execution state.

[AI HARDWARE] SOURCE: reddit.com | VEL: 96/100

Running Vision Qwen 3.8 27B on a 16GB Card, the config (45tks).

ARCH: 27B • CTX: 85K • QUANT: GGUF • STACK: PyTorch / ONNX

I am just sharing my config for Qwen 3.8 27b that fits on a 5060TI, what is cool about this is that you can even get vision! and a 85K context (I have 1.5gb of headroom for more context or a better quant) Model: IQ3_XXS-mtp from Using beellama Config used: [*] model = ..\llm-models\Qwen3.8-27B-GSQ-RCO-IQ3_XXS-mtp.gguf mmproj = ..\llm-models\mmproj-Qwen3.8-27B-BF16.gguf image-min-tokens = 256.

[AI HARDWARE] SOURCE: reddit.com | VEL: 96/100

Qwen3.8-Flash-Next on 2x3090 + DDR4, part 4: 2.2-2.5x faster prefill by kicking the expert cache off the GPU while the prompt runs

ARCH: MoE / SOTA • CTX: 8K • QUANT: UD-Q4_K_XL • STACK: llama.cpp

. 17 -> 25-29 t/s with the expert cache PR, 37-41 t/s after switching to UD-Q4_K_XL and stacking MTP on the cache, the top-k fallback that was sorting more than it needed to. This one is all about prefill, which was honestly the weak spot the whole time. 80+ seconds before the first token on an 8k prompt, and 24 minutes on a 119k one...I know . Box is still 2x 3090, dual Broadwell Xeon, llama.cpp, UD-Q4_K_XL with the Q8 MTP head on the.

[AI HARDWARE] SOURCE: reddit.com | VEL: 96/100

Qwen3.8-Flash-Next on MLX-serve, 1m context is released!

ARCH: MoE / SOTA • CTX: 1M • QUANT: Native / FP16 • STACK: MLX

Hi, I'm the co-creator of this Qwen3.8-Flash-Next engine support in MLX-serve. I've been tuning this one to run both fast, efficient and correct up 1m context using kv cache 8 bits in M5 Max 128GB. Qwen is working well at very long context as showed in the video (a snapshot at ~760k context), I let it build MLX Serve Monitor plugin that you've seen on the right side of Opencode2's app. The generation sustain through 1m context at around 40 tok/s on prose and 75 tok/s on coding.

[AI HARDWARE] SOURCE: github.com | VEL: 88/100

llama.cpp High-Performance LLM Engine: b10899

ARCH: MoE / SOTA • CTX: 128K • QUANT: Native / FP16 • STACK: llama.cpp

vulkan: small M matrix optimizations for qwen (#28457) vulkan: optimize m=1 mul_mat by swapping A/B vulkan: Improve small M perf Allow split_k with small M. Make small vs med tile selection (for coopmat2) depend on M, not just N.

[AI HARDWARE] SOURCE: arxiv.org | VEL: 88/100

Substrate-Aware AI Agents: Execution Context as a First-Class Input

ARCH: MoE / SOTA • CTX: 128K • QUANT: Native / FP16 • STACK: PyTorch / ONNX

Autonomous AI agents increasingly select actions in environments whose memory, execution-time, runtime, compute, and operational constraints determine what counts as a suitable plan. We call the absence of this execution context from an agent's planning state substrate blindness. We test this general proposition through numerical code generation, where selected implementation choices and operational consequences are directly observable.

■ SECTION IX: AUTONOMOUS AGENTS, MULTI-AGENT SWARMS & ROBOTICS

Agent Orchestration, Computer-Use Primitives, Action Models & Robotics

AGENT / FRAMEWORK	SOURCE	PROTOCOL	VELOCITY	CORE CAPABILITY DOMAIN
Enhancing Virtual Agents through SLMs and Edg	arxiv.org	PyTorch / ONNX	88/100	Embodied intelligent virtual agents are expected to operate as persistent, adaptive, and context-aware entities within complex virtual and.
Social Chain of Thought: A Multi-Agent Archit	arxiv.org	PyTorch / ONNX	88/100	Medical diagnostic reasoning is a high-impact use case for LLMs that carries significant implications for the health and.
When LLM Agents Negotiate: Private Informatio	arxiv.org	PyTorch / ONNX	88/100	As LLM agents move from decision support to autonomous procurement, firms need to know whether delegated negotiators create.
Research acceleration: The view inside OpenAI	simonwillison.ne	PyTorch / ONNX	96/100	Research acceleration: The view inside OpenAI Apparently today is RSI day at OpenAI, for Recursive Self-Improvement - I.
LangChain AI Application Framework: langchain	github.com	PyTorch / ONNX	88/100	LangChain is a framework for building agents and LLM-powered applications. It helps you chain together interoperable components and.
Scanning the Harness: An Empirical Study of S	arxiv.org	PyTorch / ONNX	88/100	AI coding agents such as Claude Code, Cursor, GitHub Copilot, and OpenAI Codex are configured through artifacts developers.

[AGENT SYSTEM] SOURCE: [arxiv.org](#) | VEL: 88/100

Enhancing Virtual Agents through SLMs and Edge-Computing: An Exploratory Evaluation of Think and Memory Processes

ARCH: MoE / SOTA • CTX: 128K • QUANT: Native / FP16 • STACK: PyTorch / ONNX

Embodied intelligent virtual agents are expected to operate as persistent, adaptive, and context-aware entities within complex virtual and Metaverse worlds. However, implementing cognitively capable agents in such environments is conceptually and technologically challenging.

[AGENT SYSTEM] SOURCE: [arxiv.org](#) | VEL: 88/100

Social Chain of Thought: A Multi-Agent Architecture Grounded in Medical Differential Diagnosis Methodology

ARCH: MoE / SOTA • CTX: 128K • QUANT: Native / FP16 • STACK: PyTorch / ONNX

Medical diagnostic reasoning is a high-impact use case for LLMs that carries significant implications for the health and wellbeing of users. When OpenAI (2026) reports that more than 5% of ChatGPT messages globally are healthcare-related, the transparency of these systems becomes a serious design concern. This is especially true for complex cases, where differential diagnosis often requires integrating multiple forms of specialist reasoning.

[AGENT SYSTEM] SOURCE: [arxiv.org](#) | VEL: 88/100

When LLM Agents Negotiate: Private Information and Dynamic Bargaining in Supply Chains

ARCH: MoE / SOTA • CTX: 128K • QUANT: Native / FP16 • STACK: PyTorch / ONNX

As LLM agents move from decision support to autonomous procurement, firms need to know whether delegated negotiators create value, divide it predictably, and avoid money-losing contracts. We study this in a canonical supply chain bargaining problem: a buyer with private demand information negotiates a quantity-payment contract with an uninformed seller. We benchmark nine LLMs from OpenAI, Google, and Alibaba against a validated Perfect Bayesian Equilibrium across 9,840 LLM-to-LLM negotiations.

[AGENT SYSTEM] SOURCE: [simonwillison.net](#) | VEL: 96/100

Research acceleration: The view inside OpenAI

ARCH: MoE / SOTA • CTX: 128K • QUANT: Native / FP16 • STACK: PyTorch / ONNX

Research acceleration: The view inside OpenAI Apparently today is RSI day at OpenAI, for Recursive Self-Improvement - I think it's their new AGI. Both this piece and the new essay An Alien Mind (by Chief Scientist Jakub Pachocki) talk about it, and this one doesn't even bother to expand the acronym. Included are details on how OpenAI's own research team are using coding agents.

[AGENT SYSTEM] SOURCE: [github.com](#) | VEL: 88/100

LangChain AI Application Framework: langchain-anthropic==1.7.2

ARCH: MoE / SOTA • CTX: 128K • QUANT: Native / FP16 • STACK: PyTorch / ONNX

LangChain is a framework for building agents and LLM-powered applications. It helps you chain together interoperable components and third-party integrations to simplify AI application development — all while future-proofing decisions as the underlying technology evolves. > [!TIP] > Just getting started? Check out **Deep Agents** — a higher-level package built on LangChain for agents that have built-in capabilities for common usage patterns such as planning, subagents, file system usage, and more. If you're looking for more advanced customization or agent orchestration, check out LangGraph, our framework for building controllable agent workflows.

■ SECTION X: GLOBAL AI COMMUNITY WIRE & OVERFLOW DIGEST

High-velocity AI stories from today's sweep that didn't fit earlier sections

[OVERFLOW] [SOURCE: arxiv.org](#) | [VEL: 88/100](#)

Robust for the Wrong Reasons: The Representational Geometry of LLM Robustness to Science Skepticism

ARCH: 8B • **CTX:** 128K • **QUANT:** Native / FP16 • **STACK:** PyTorch / ONNX

Large language models (LLMs) are increasingly consulted on contested scientific questions, raising the concern that they will sycophantically retreat from established consensus when a user signals doubt -- drifting toward a false balance that treats settled science as one view among several. We test this across three open instruction-tuned models (Llama-3.1-8B, Qwen2.5-7B, Mistral-7B), three consensus-science domains (climate, vaccines, evolution), and single- and multi-turn settings, combining.

[OVERFLOW] [SOURCE: arxiv.org](#) | [VEL: 88/100](#)

The Viola Architecture for Efficient Small Language Models

ARCH: MoE / SOTA • **CTX:** 128K • **QUANT:** Native / FP16 • **STACK:** PyTorch / ONNX

We present Viola, a fully original Small Language Model (SLM) architecture built from first principles, sharing no structural lineage with any existing model family including GPT, LLaMA, Mistral, or Falcon.

[OVERFLOW] [SOURCE: arxiv.org](#) | [VEL: 88/100](#)

Research Entity Extraction and Topic Detection from UKRI Grant Proposals

ARCH: MoE / SOTA • **CTX:** 128K • **QUANT:** Native / FP16 • **STACK:** PyTorch / ONNX

This paper presents preliminary findings from a UKRI-funded Metascience project comparing three LLM-based approaches, GPT-4o, Mistral, and a bespoke algorithm, DSIT-Taxonomies, for extracting and classifying research entities from funding proposals. Our project "Tracking Stars and Unicorns" aims to identify early signals of emerging research areas to inform public investment.

[OVERFLOW] [SOURCE: arxiv.org](#) | [VEL: 88/100](#)

The Truth Stays in the Family: Enhancing Contextual Grounding via Inherited Truthful Heads in Model Lineages

ARCH: MoE / SOTA • **CTX:** 128K • **QUANT:** Native / FP16 • **STACK:** PyTorch / ONNX

Recent advances in large language models (LLMs) have produced many specialized multimodal LLMs (MLLMs) that share common foundational LLMs, forming distinct model lineages. It remains unclear whether a fundamental behavioral link exists between the foundational LLMs and downstream variants. We investigate this question by quantifying head-level context-truthfulness scores. Across diverse LLM and MLLM lineages, including Vicuna-, Qwen2.5-, LLaMA2-, and Mistral-based models, we find that Truth.

[OVERFLOW] [SOURCE: simonwillison.net](#) | [VEL: 88/100](#)

Codex bundles LibreOffice

ARCH: MoE / SOTA • **CTX:** 128K • **QUANT:** Native / FP16 • **STACK:** PyTorch / ONNX

I was poking around in my ~/.cache/ folder using OmniDiskSweeper when I spotted something interesting. The OpenAI Codex desktop app (since rebranded to just ChatGPT) has 1.7GB of stuff in there in a folder called codex-primary-runtime, including a full Python installation, a full Node.js installation, and native binaries for Poppler, git, and the LibreOffice open source office suite (which forked from OpenOffice.org in 2010): The ~/.cache/codex-runtimes/codex-primary-runtime/plugins/openai-prima.

[OVERFLOW] [SOURCE: aws.amazon.com](#) | [VEL: 88/100](#)

Deploying Qwen3.8-2.4T-A95B on Amazon SageMaker HyperPod with vLLM

ARCH: MoE / SOTA • **CTX:** 262K • **QUANT:** Native / FP16 • **STACK:** vLLM

On August 12, 2026, Alibaba's Qwen team released Qwen3.8-2.4T-A95B. This is the first time a Qwen-Max-class model has been made available as open weights. With 2.4 trillion total parameters (95 billion activated per token), a hybrid linear-plus-full-attention architecture, and native context up to 262K tokens (extensible to 1M), Qwen3.8 targets the most demanding agentic and reasoning workloads. These include multi-step coding, long-horizon planning, and autonomous tool use.

[OVERFLOW] [SOURCE: vttresearch.com](#) | [VEL: 88/100](#)

SemiQon's cryogenic chip technology for quantum computing and space applications receives award from EARTO

ARCH: MoE / SOTA • **CTX:** 128K • **QUANT:** Native / FP16 • **STACK:** PyTorch / ONNX

EARTO, the organisation of the European Research and Technology Organisations, awarded SemiQon and VTT first prize in the "Impact Expected" category on 14 October 2025 in Brussels for a pioneering cryogenic CMOS (complementary metal-oxide semiconductor) chip innovation. The solution enables the full capacity of advanced CMOS functionalities at cryogenic temperatures, thereby unlocking new possibilities for quantum computing and space applications.

[OVERFLOW] [SOURCE: pandaily.com](#) | [VEL: 88/100](#)

DeepSeek Ships V4.1 Flash GA With Causal-Encoder-Decoder MoE as V4 Pro Retires

ARCH: MOE • **CTX:** 128K • **QUANT:** Native / FP16 • **STACK:** PyTorch / ONNX

DeepSeek has released DeepSeek V4.1 Flash for general availability, promoting the model from a short limited beta into a production SKU on a new Causal-Encoder-Decoder Mixture-of-Experts architecture. The company presents V4.1 Flash as the smallest member of that structure family and says it now surpasses DeepSeek V4 Pro on capability, cost, speed and end-to-end task time, clearing the path for an orderly Pro retirement later this week. Architecturally, V4.1 Flash is a 552-billion-parameter.

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